Zhejiang University-University of Illinois Urbana-Champaign Institute

(ZJU-UIUC INSTITUTE)

Electronics and Computer Engineering Curriculum

(Updated in August 2025)

1. Program Educational Objectives

Electronics and Computer engineering focuses on the development of vital computing technologies, ranging from chips to computers to networks to programming tools to key algorithms for building exciting applications. Fundamentally, computer engineering addresses the problem of building scalable, trustworthy computing systems, and the faculty's interests span a broad spectrum of issues pertinent to this theme. Computer engineering has taken the lead in revolutionizing many science and engineering disciplines with parallel computing, from chips to clouds to planet-scale critical infrastructures, and has defined new standards of security, privacy, and dependability for systems ranging from small circuits to the electric power grids of many nations. Students need a broad and sound set of mathematical and computing skills and are well-served by a flexible curriculum that enables them to pursue topics of interest among the many sub-disciplines in computing.

2. Graduation Requirement

The curriculum leading to the Bachelor of Engineering from ZJUI requires 137 hours and is organized into required courses and elective courses.

3. Required Courses

3.1 Orientation and Professional Development

These courses introduce the opportunities and resources our institute and curriculum can offer you as you work to achieve your career goals. They also provide the skills to work effectively and successfully in the engineering profession.

Course Code	Course Name	Credit
ENG 100	Engineering Orientation	1

3.2 Foundational Mathematics and Science

These courses stress the basic mathematical and scientific principles upon which the engineering discipline is based.

Course Code	Course Name	2020	2021-
MATH 221	Calculus I	4	4
MATH 231	Calculus II	3	3
MATH 241	Calculus III	4	4
MATH 257	Linear Algebra with Computational Application		3
MATH 285	Introduction to Differential Equation		3
MATH 286	Introduction to Differential Equation Plus	4	
PHYS 211	University Physics: Mechanics	4	4
PHYS 212	University Physics: Elec & Mag	4	4
PHYS 213	Univ Physics: Thermal Physics	2	2
PHYS 214	Univ Physics: Quantum Physics	2	2
CHEM 102*	General Chemistry I	3	
CHEM 103*	General Chemistry Lab I	1	
	Total	31	29

**Elective for CompEs on the Fall 2021 or later catalog year.*

3.3 Technical Core

These courses stress fundamental concepts and basic laboratory techniques that comprise the common intellectual understanding of computer engineering.

Course Code	Course Name	2020-
ECE 110	Introduction to Electronics	3
ECE 120	Introduction to Computing	4
ECE 210	Analog Signal Processing	4
ECE 220	Computer Systems & Programming	4
MATH 213	Basic Discrete Mathematics	3
CS 225	Data Structures	4
ECE 313	Probability with Engineering Application	3
ECE 374	Introduction to Algorithms & Models of Computation	4
ECE 385	Digital Systems Laboratory	3
ECE 445	Senior Design Project Lab	4
	Total	36

4. Elective Courses

4.1 Technical Electives

These courses stress cross-disciplinary study, the rigorous analysis and design principles practiced in the major sub-disciplines of computer engineering, for example cyber physical systems; foundations and theory; software and languages; algorithms and mathematical tools; trust, reliability, security; networking, mobile and distributed computing; big data analytics and systems; artificial intelligence, robotics, cybernetics, etc.

Students must complete a minimum of total required hours on technical electives, which include at least:

1) 2 courses **EE Foundations Courses**, see section 4.1.1 for detail.

2)	2 courses	**Advanced Computing Electives**, see section 4.1.2 for detail.	
3)	Others	See section 4.1.3 for detail	
		Total required	29

4.1.1 EE Foundation Courses

Select 2 of 6 EE Foundation Courses:

Course Code	Course Name	Credits
ECE 310	Digital Signal Processing	3
ECE 330	Power Circuits & Electromechanics	3
ECE 329	Fields and Waves I	3
ECE 340	Semiconductor Electronics	3
ECE 461	Digital Communications	3
ECE 486	Control Systems	4

4.1.2 Advanced Computing Electives

Following CS and ECE courses are listed as advanced computing electives. Some ECE ones are cross listed with CS. Select two of them.

Course Code	Course Name	Credits
CS 357	Numerical Methods I	3
CS 411	Database Systems	3
CS 412	Introduction to Data Mining	3
CS 414	Multimedia Systems	3
CS 418	Interactive Computer Graphics	3
CS 419	Production Computer Graphics	3
CS 420	Parallel Programming: Scientists and Engineers	3
CS 421	Programming Languages & Compilers	3
CS 423	Operating Systems Design	3
CS 424	Real-Time Systems	3

CS 425	Distributed Systems	3
CS 426	Compiler Construction	3
CS 431	Embedded Systems	3
CS 436	Computer Networking Laboratory	3
CS 437	Topics in Internet of Things	3
CS 438	Communication Networks	3
CS 440	Artificial Intelligence	3
CS 441	Applied Machine Learning	3
CS 444	Deep Learning for Computer Vision	3
CS 446	Machine Learning	3
CS 450	Numerical Analysis	3
CS 461	Computer Security I	3
CS 473	Algorithms	4
CS 475	Formal Models of Computation	3
CS 476	Program Verification	3
CS 477	Formal Software Development Methods	3
CS 483	Applied Parallel Programming	3
ECE 407/CS 407	Cryptography	3
ECE 408/CS 483	Applied Parallel Programming	4
ECE 411	Computer Organization & Design	4
ECE 412	Microcomputer Laboratory	3
ECE 419/CS 460	Computer Security Laboratory	3
ECE 422/CS 461	Computer Security I	4
ECE 424/CS 463	Computer Security II	3
ECE 425	Intro to VLSI System Design	3
ECE 428/CS 425	Distributed Systems	3
ECE 434	Mobil Computing and Application	3
ECE 435/CS 436	Computer Networking Laboratory	3
ECE 438/CS 438	Communication Networks	3
ECE 439/CS 439	Wireless Networks	3
ECE 448/CS 440	Artificial Intelligence	3
ECE 449/CS 446	Machine Learning	3
ECE 462	Logic Synthesis	3
ECE 470	Introduction to Robotics	4
ECE 478/CS 477	Formal Software Development Methods	3

ECE 484	Principles of Safe Autonomy	4
ECE 491/CS 450	Numerical Analysis	3
ECE 492/CS 420	Parallel Programming: Scientists and Engineers	3

4.1.3 Other Electives

Course Code	Course Name	Credits
CEE 310	Transportation Engineering	3
ME 200	Thermodynamics	3
ME 320	Heat Transfer	4
ME 371	Mechanical Design II	3
ME 400	Energy Conversion Systems	3 or 4
ME 412	Numerical Thermo-Fluid Mechs	2 to 4
ME 472	Introduction to Tribology	3
ME 485	MEMS Devices & Systems	3
ME 498VS	Vacuum Engineering in Semiconductor Industry	3
TAM 211	Statics	3
TAM 212	Introductory Dynamics	3
TAM 251	Introductory Solid Mechanics	3
TAM 335	Introductory Fluid Mechanics	4
CS 101	Introduction to Computing: Engineering & Science	3
CS 225	Data Structures	4
CS 357	Numerical Methods I	3
CS 412	Introduction to Data Mining	3 or 4
CS 442	Trustworthy machine learning	3
ECE 310	Digital Signal Processing	3
ECE 311	Digital Signal Processing Lab	1
ECE 314	Probability in Engineering Lab	1
ECE 329	Fields and Waves I	3

ECE 340	Semiconductor Electronics	3
ECE 342	Electronic Circuits	3
ECE 343	Electronic Circuits Laboratory	1
ECE 350	Fields and Waves II	3
ECE 364	Programing Methods for Machine Learning	3
ECE 365	Data Science and Engineering	3
ECE 438	Communication Networks	3 or 4
ECE 441	Physics & Modeling Semiconductor Devices	3
ECE 448	Artificial Intelligence	3 or 4
ECE 453	Wireless Communication Systems	4
ECE 457	Microwave Devices & Circuits	3
ECE 459	Communications Systems	3
ECE 464	Power Electronics	3
ECE 469	Power Electronics Laboratory	2
ECE 470	Introduction to Robotics	4
ECE 471	Data Science Analytics using Probabilistic Graph Models	3
ECE 476	Power System Analysis	3
ECE 485	MEMS Devices & Systems	3
ECE 486	Control Systems	4
ECE 491	Numerical Analysis	3 or 4
ECE 498MF	Mathematical Foundations for Machine Learning	3
ECE 498EAI	Embodied Artificial Intelligence	3
ABE 436	Renewable Energy Systems	3
CHEM 104	General Chemistry II	3
CHEM 105	General Chemistry Lab II	1

4.2 Liberal Education

The liberal education courses develop students' understanding of human culture and society, build skills of inquiry and critical thinking, and lay a foundation for civic engagement and lifelong learning. To satisfy the General Education requirements, students must take all courses for grade and complete courses based on the table below:

Category	Credit
1) Humanities & Arts (2 courses)	6
2) Social & Behavioral Sciences (2 courses)	6
3) Culture Studies	6
Western/Comparative Culture(s) (1 course) or	
Non-Western Culture(s) (1 course)	
Total required	**18**

To meet the General Education requirements, students need to select courses that fulfill these four subcategories and accumulate a total of 18 credits. Properly chosen courses in the Social and Behavioral Sciences and Humanities & Arts categories can also satisfy the Western and non-Western culture requirements, ensuring that students receive a comprehensive education.

4.3 Free Electives

These unrestricted electives give the student the opportunity to explore any intellectual area of unique interest. This freedom plays a critical role in helping students to define research specialties. Students are encouraged to take cross-discipline courses as free electives.

Free electives	2020	2021-2022	2023
Total	0	2	1

5. ZJU Required Liberal Education

These courses introduce Chinese modern history, social development, government policies, etc., help students to improve their English and maintain a healthy lifestyle.

International students need to complete Chinese language study and Chinese society to fulfill the graduate requirement.

Course Code	Course Name	2020 Credits	2021-2022 Credits	2023- Credits
CHIN 1001	Chinese I	4	4	4
CHIN 1002	Chinese II	5	5	4
CHIN 1003	Chinese III	4	4	4
CHIN 1004	Chinese IV	4	4	4
CHIN1005	International Chinese Language Proficiency and Testing			2
CULT 2001	China Survey	3	3	3
ST 4001	Senior Thesis		2	2
	Total	20	22	23